Jeremy Goldwasser

Website · Email · Linkedin · Google Scholar

Education

UC Berkeley, Berkeley, CA — Aug 2021 - May 2026 (Expected)
Ph.D. in Statistics; Advised by Ryan Tibshirani and Giles Hooker — GPA: 3.97

Yale University, New Haven, CT — Aug 2017 - May 2021
B.S. in Statistics and Data Science — GPA: 3.73

Internship Experience

Apple — May - Aug 2024
Research Intern, Vision Products Group — Sunnyvale, CA

- Developed novel generative AI method for counterfactual image explanations.
- Tool is currently used to provide previously unattainable insights on failure modes of FaceID.

Genentech — May - July 2023
Data and Statistical Sciences Intern — South San Francisco, CA

- Produced confidence intervals for AI pathology models using conformal prediction.
- Calibrated cell-type classifier, quantifying uncertainty and improving test accuracy by 3%.

Voca.ai (acquired by Snapchat) — June - Aug 2019
Machine Learning Intern — Herzliya, Israel

- Trained Transformers for Automatic Speech Recognition and Named Entity Recognition.

Publications

Unifying Image Counterfactuals and Feature Attributions with Latent-Space Adversarial Attacks
ICML 2025, Actionable Interpretability Workshop – First author

Statistical Significance of Feature Importance Rankings
Conference on Uncertainty in AI (UAI) 2025 – First author

Gaussian Rank Verification
Stat, 2025 – First author

Stabilizing Estimates of Shapley Values with Control Variates
World Conference on Explainability in AI (XAI) 2024 – First author

Ascle: A Python Natural Language Processing Toolkit for Medical Text Generation
Journal of American Medical Informatics Association (JAMIA), 2024

Forest Fire Clustering for Single-Cell Sequencing Combines Iterative Label Propagation with Parallelized Monte Carlo Simulations
Nature Communications, 2022 – Second author

Neural NLP for Unstructured Data in Electronic Health Records: A Review
Computer Science Review, 2022 – Third author

Submissions

Challenges in Real-Time Estimation of Changing Epidemic Severity Rates
In Review, PLOS Computational Biology – First author, 2024

Ongoing Projects

- Statistical ML algorithms to better track mortality risk in epidemics like COVID-19.
- Attention-based interpretability of Vision Transformers for AI cancer pathology.
- Online learning for ensembling time series forecasters with uncertainty quantification.

Misc Projects

AI Meets DNA Methylation May 2024
- Used interpretable ML to identify methylation sites that regulate CD55 gene expression.

Predicting Peptide-MHC Binding Affinity in SARS-CoV-2 April 2020
- Designed neural architecture to predict which viral peptides bind with immune cells.

Skills

Coursework: Deep Learning, Optimization, Causal Inference, AI in Biology
Programming: Python, R, LaTeX
Languages: Spanish (fluent), Hebrew (basic)

Awards

U.S. Civic Digital Fellowship (Declined) 2021
Yale College Dean's Research Fellowship, Morse Richter Fellowship 2020
Yale Creative and Performing Arts Grant 2020

Invited Talks

- **"Estimating Epidemic Severity Rates."** UCSF MINDSCAPE, February 2025.
- **"Estimating Epidemic Severity Rates."** Delphi Group, January 2025.
- **"Estimation of Time-Varying Severity Rates from Aggregate Data: Pitfalls and New Ideas."** California Department of Public Health, September 2024.
- **"Estimation of Time-Varying Severity Rates from Aggregate Data: Pitfalls and New Ideas."** Centers for Disease Control and Prevention (CDC), September 2024.
- **"Stabilizing Estimates of Shapley Values with Control Variates."** 2nd World Conference on Explainable AI, Malta, July 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Apple Data Analytics and Quality Group, June 2024.
- **"Stable Feature Rankings & Time Series Deconvolution."** Genentech gRED Computational Sciences, February 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Center for Human-Compatible AI, January 2024.
- **"Provably Stable Feature Rankings with SHAP and LIME."** Berkeley Statistics Student Seminar, January 2024.
- **"Estimating Case Fatality Rate via Convolutional Modeling."** Delphi Group, April 2023.